

Project Report: Import Data using Transform Maps (Spreadsheet)

Team ID: SWTID-2026-8759 · Date: 01-10-2026

1. Introduction

1.1 Project Overview

This project imports bulk employee data from an Excel spreadsheet into ServiceNow using Import Sets and Transform Maps. The data is loaded into a staging table, mapped to a custom Employee Test table, protected from duplicates with Coalesce on Employee ID, and shown in three reports on the Employee Analytics Dashboard.

1.2 Purpose

The purpose is to give beginners an end-to-end view of the ServiceNow data import process: data loading, field mapping, transformation, result verification, and reporting. It also shows how repeated imports update existing records instead of creating duplicates.

2. Ideation Phase

2.1 Problem Statement

Employee data arrives in Excel files and is slow and error-prone to enter into ServiceNow by hand. Repeated imports of the same data risk duplicate records, and admins and HR teams have no quick view of the data once it is loaded.

2.2 Empathy Map Canvas

Quadrant	Admin / HR user
Says	"Entering these employees one by one takes too long"
Thinks	"What if I load the same file twice and create duplicates?"
Does	Receives Excel files, copies rows into forms, checks for duplicates by eye

Feels

Stressed about errors, frustrated by repeated work, relieved when data loads cleanly

2.3 Brainstorming

Idea	Outcome
Enter each employee manually	Rejected: slow and error-prone
Write a custom script or integration	Rejected: needs developer skills, too heavy for beginners
Load the spreadsheet with Import Set and Transform Map	Selected: low-code, repeatable, no scripting
Use Coalesce on Employee ID	Selected: stops duplicates and updates changed records
Add reports and a dashboard	Selected: gives HR teams visibility into the data

3. Requirement Analysis

3.1 Customer Journey Map

Stage	Action	Feeling	Opportunity
Receive data	Employee data arrives as an Excel file	Neutral	Standardise the five columns
Load	Upload the file through Load Data	Hopeful	One guided screen creates the staging table
Map	Match source fields to target fields	Careful	Auto Map and Mapping Assist save time
Transform	Run the transform and check the result	Relieved on Success	Transform history shows inserts, updates and errors
Re-import	Load changed data again	Worried about duplicates	Coalesce updates or ignores existing rows

Review

Open reports and
the dashboard

Confident

One view of
department and
location data

3.2 Solution Requirement

The solution has eight functional requirements: source data preparation, target table creation, data loading, field mapping, transformation and validation, duplicate prevention with Coalesce, reporting, and a dashboard with access control. Six non-functional requirements apply: usability, security, reliability, performance, availability and scalability.

3.3 Data Flow Diagram

1. User prepares employee data in Google Sheets and downloads Sample Spreadsheet.xlsx
2. Load Data uploads the file into the staging table u_employee_import
3. The Transform Map Sample Spreadsheet Import maps source fields to u_employee_test
4. Coalesce on Employee ID inserts new rows, updates changed rows and ignores unchanged rows
5. Employee Test holds the final records
6. Three reports read from Employee Test and appear on the Employee Analytics Dashboard

3.4 Technology Stack

Layer	Technology
User interface	ServiceNow Now Platform UI in a web browser
Data loading	Import Sets / Load Data
Mapping and logic	Transform Map with Coalesce
Database	ServiceNow custom tables u_employee_import and u_employee_test
Reporting	ServiceNow Reports and Platform Analytics Dashboards
Source data	Google Sheets and Excel (.xlsx)
Infrastructure	ServiceNow cloud instance

4. Project Design

4.1 Problem Solution Fit

The customers are ServiceNow admins and HR teams who receive employee data in spreadsheets. Their problem is manual entry and duplicate risk. Import Sets with a Transform Map and Coalesce fit how they already work, because the data stays in Excel and only the loading step changes.

4.2 Proposed Solution

Load the Excel file into a staging table, map it to the Employee Test table with a Transform Map, enable Coalesce on Employee ID, and show the data in three reports on a dashboard. The novelty is combining import, duplicate control and reporting in one low-code flow.

4.3 Solution Architecture

The architecture has a local side (Google Sheets and the .xlsx file) and a ServiceNow cloud side (Import Set staging table, Transform Map, Employee Test table, reports and dashboard). The pa_dashboards ACL controls dashboard creation.

5. Project Planning & Scheduling

5.1 Project Planning

Sprint	Story Points	Duration	Start	Planned End	Focus
Sprint-1	12	6 Days	19 Jan 2026	24 Jan 2026	Data preparation, tables, import, Transform Map, Coalesce
Sprint-2	14	6 Days	26 Jan 2026	31 Jan 2026	Re-import tests, reports, dashboard, access control

Average velocity is 13 story points per sprint, about 2.17 story points per day.

6. Functional and Performance Testing

6.1 Performance Testing

Test	Rows	Inserts	Updates	Ignored	Errors	Result
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First import and transform	15	15	0	0	0	Success, about 1.4 seconds to load
Updated file (2 changed rows, 2 new rows)	4	2	2	0	0	Success, about 1 second to transform
Same file again	4	0	0	4	0	Success, 0 seconds, no duplicates

7. Results

7.1 Output Screenshots

The final Employee Test table holds 17 employees. Department split is ServiceNow 11, Salesforce 4 and AIML 2. Location split is Hyderabad 16 and Chennai 1.

Add these screenshots here:

- Employee Test table after the first import (15 records)
- Transform history for the three imports
- Employee Test table after the update (17 records)
- Employees by Department, Employees by Location and Employee List reports
- Employee Analytics Dashboard

8. Advantages & Disadvantages

Advantages

- Low-code: no scripting needed
- Coalesce prevents duplicate records and updates changed ones
- Staging table separates loading from transformation, so errors are caught before the target table
- Reports and dashboard give instant visibility into the data

Disadvantages

- Manual steps: file download, upload and Run Transform each time
- Requires admin access to create tables, maps and ACL changes
- Only the Employee ID field is coalesced, so a wrongly typed ID creates a new record

- All fields are String type, so there is no validation of email or other formats

9. Conclusion

The project loaded employee data from a spreadsheet into ServiceNow with Import Sets and a Transform Map, and Coalesce on Employee ID kept repeated imports clean: new IDs were inserted, changed rows were updated, and identical rows were ignored. Three reports on one dashboard gave a clear view of the data.

10. Future Scope

- Schedule imports automatically from a data source instead of manual file loads
- Add validation for email and other fields with transform scripts
- Add reports for newly added employees, updated records and import success and error counts
- Reuse the same approach for other tables such as users and assets

11. Appendix

- **Source Code:** none, the solution uses ServiceNow configuration only
- **Dataset Link:** Sample Spreadsheet.xlsx (add link)
- **GitHub & Project Demo Link:**
(<https://github.com/praveenhari402/servicenow-transform-map-project>)

Project Documentation: Import Data using Transform Maps (Spreadsheet)

This project runs on ServiceNow, so the MERN parts of the format are mapped to their ServiceNow equivalents and marked where they do not apply.

1. Introduction

- **Project Title:** Import Data using Transform Maps (Spreadsheet)
- **Team ID:** SWTID-2026-8759
- **Team Members:** Praveen (role: lead).

2. Project Overview

- **Purpose:** Import bulk employee data from an Excel spreadsheet into ServiceNow accurately, avoid duplicate records on repeat imports, and show the data in reports and a dashboard
- **Features:**
 - o Custom table Employee Test (u_employee_test) with five String fields
 - o Spreadsheet load into the staging table u_employee_import
 - o Transform Map with Auto Map and Mapping Assist field mapping
 - o Coalesce on Employee ID: new IDs inserted, changed rows updated, identical rows ignored
 - o Three reports and the Employee Analytics Dashboard

3. Architecture

Layer	MERN format	This project
Frontend	React	ServiceNow Now Platform UI: forms, lists, Load Data screen, reports and dashboards in the browser
Backend	Node.js and Express.js	ServiceNow platform: Import Set processor and Transform Map engine
Database	MongoDB	ServiceNow relational database with two custom tables

Database schema

Table	Fields
u_employee_import (staging)	u_employee_id, u_name, u_email, u_department, u_location, plus system columns
u_employee_test (target)	Employee ID, Employee Name, Email, Department, Location (all String)

The Transform Map maps u_name to u_employee_name, and the other four fields to their matching target fields. Coalesce is true on u_employee_id.

4. Setup Instructions

Prerequisites

- A ServiceNow instance with admin access
- A Google account (Google Sheets) or Excel to prepare the data

Installation (no repository to clone, everything is configured in the instance)

1. Create the employee data in a spreadsheet and download it as Sample Spreadsheet.xlsx
2. Create the table Employee Test (u_employee_test) and add the five String fields
3. Open Load Data, create the table Employee Import (u_employee_import) from the file
4. Create the Transform Map Sample Spreadsheet Import from the load result and map the fields
5. Set Coalesce to true on the Employee ID field map
6. Open the ACL pa_dashboards create operation and enable Admin overrides if the dashboard form is blocked

5. Folder Structure

There is no client or server code folder. The project is made of configuration records in the instance:

- **Client:** none. The UI is the built-in ServiceNow interface
- **Server:** none. Logic is the Import Set, the Transform Map and its field maps

6. Running the Application

There are no start commands. To run an import:

1. Open Load Data and select the existing table Employee Import
2. Choose the file, set Sheet number 1 and Header row 1, then submit
3. Click Run Transform, select Sample Spreadsheet Import and click Transform

4. Open Transform History to see inserts, updates, ignored rows and errors

7. API Documentation

The project exposes no custom endpoints. All interaction is through ServiceNow UI modules.

8. Authentication

Users sign in to the ServiceNow instance. Access is role-based: dashboard creation is controlled by the pa_dashboards ACL (create operation, Admin overrides true), and the dashboard can be shared with specific users, groups and roles from its Share option.

9. User Interface

Add screenshots of:

- Employee Test table form and list
- Load Data screen
- Transform Map field maps
- Three reports
- Employee Analytics Dashboard

10. Testing

Testing was done by running three imports and checking the Transform History.

Test	Rows	Inserts	Updates	Ignored	Errors
First import	15	15	0	0	0
Updated file (2 changed rows, 2 new rows)	4	2	2	0	0
Same file again	4	0	0	4	0

The final table holds 17 employees, and the list, pie and bar reports were checked against it.

11. Screenshots or Demo

<https://github.com/praveenhari402/service-now-transform-map-project>

12. Known Issues

- The Department value appears as "Servicenow" (lowercase n) on some rows after the update import, which splits the department count if it is not corrected
- The Email field is a plain String, so formats are not validated
- Imports need manual steps each time (upload and Run Transform)

13. Future Enhancements

- Scheduled imports from a data source
- Field validation scripts for email and department values
- Reports for newly added records, updated records and import success and error counts
- Reuse the same flow for other tables such as users and assets